

Supplementary Material for “Health-informed Optimal Vehicle-to-Grid Dispatch with Thermal Runaway Mitigation via Digital Twin”

Chao Lei, *Member, IEEE*, Jessie Ma, *Senior Member, IEEE*, and Yu Christine Chen, *Member, IEEE*

I. DERIVATIONS OF (5)

Substitute (1) into (2) and rearrange the resultant for $j = 0, 1, \dots, n-1$ to yield

$$\begin{bmatrix} \hat{T}_{c,i,j+1}^t \\ \hat{T}_{s,i,j+1}^t \end{bmatrix} = \begin{bmatrix} 1 - \omega_1 & \omega_1 \\ \omega_2 & 1 - \omega_3 \end{bmatrix} \begin{bmatrix} T_{c,i,j}^t \\ T_{s,i,j}^t \end{bmatrix} + \begin{bmatrix} \frac{\Delta h}{C_{c,i}} H_{e,i}^t \\ \frac{T_{amb,i} \Delta h}{C_{s,i} R_{s,i}} \end{bmatrix}, \quad (I-1)$$

$$\omega_1 = \frac{\Delta h}{C_{c,i} R_{c,i}}, \quad \omega_2 = \frac{\Delta h}{C_{s,i} R_{c,i}}, \quad \omega_3 = \frac{\Delta h}{C_{s,i}} \left(\frac{1}{R_{s,i}} + \frac{1}{R_{c,i}} \right).$$

Further substitute (1) into the corrector step in (3) to get

$$\begin{bmatrix} T_{c,i,j+1}^t \\ T_{s,i,j+1}^t \end{bmatrix} = \begin{bmatrix} 1 - 0.5\omega_1 & 0.5\omega_1 \\ 0.5\omega_2 & 1 - 0.5\omega_3 \end{bmatrix} \begin{bmatrix} T_{c,i,j}^t \\ T_{s,i,j}^t \end{bmatrix} + \begin{bmatrix} -0.5\omega_1 & 0.5\omega_1 \\ 0.5\omega_2 & 0.5\omega_3 \end{bmatrix} \begin{bmatrix} \hat{T}_{c,i,j+1}^t \\ \hat{T}_{s,i,j+1}^t \end{bmatrix} + \begin{bmatrix} \frac{\Delta h}{C_{c,i}} H_{e,i}^t \\ \frac{T_{amb,i} \Delta h}{C_{s,i} R_{s,i}} \end{bmatrix}, \quad (I-2)$$

Next, substitute (I-1) into (I-2) to get

$$\begin{bmatrix} T_{c,i,j+1}^t \\ T_{s,i,j+1}^t \end{bmatrix} = A_1 \cdot \begin{bmatrix} T_{c,i,j}^t \\ T_{s,i,j}^t \end{bmatrix} + A_2 \cdot H_{e,i}^t + A_3, \quad j = 0, 1, 2, \dots, n-1, \quad (I-3)$$

where

$$A_1 = \begin{bmatrix} 0.5\omega_1^2 - \omega_1 + 0.5\omega_1\omega_2 + 1 & -0.5\omega_1^2 + \omega_1 - 0.5\omega_1\omega_3 \\ \omega_2 - 0.5\omega_1\omega_2 + 0.5\omega_2\omega_3 & 0.5\omega_1\omega_2 - 0.5\omega_3^2 + 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} (1 - 0.5\omega_1) \frac{\Delta h}{C_{c,i}} \\ 0.5\omega_2 \frac{\Delta h}{C_{c,i}} \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0.5\omega_1 \frac{T_{amb,i} \Delta h}{C_{s,i} R_{s,i}} \\ (1 + 0.5\omega_3) \frac{T_{amb,i} \Delta h}{C_{s,i} R_{s,i}} \end{bmatrix}.$$

With boundary conditions specified in (4), the recurrence relation in (I-3) can be evaluated in closed form as (5).

II. DERIVATION OF (10E)

Recall that in the final layer of the ICNN for EV i , $z_{i,N_I}^{t+1} = [\Delta\theta_i^{t+1}, \Delta B_i^{t+1}]^T$. Define $f := (\gamma_i^{t+1} + 1)^2$ and $g := (\gamma_i^{t+1} - 1)^2 + (2\|z_{i,N_I}^{t+1}\|_2)^2$. Then

$$f - g = 4\gamma_i^{t+1} - 4\|z_{i,N_I}^{t+1}\|_2^2 \geq 0, \quad (II-4)$$

where the inequality holds because $\gamma_i^{t+1} \geq \|z_{i,N_I}^{t+1}\|_2^2 = (\Delta\theta_i^{t+1})^2 + (\Delta B_i^{t+1})^2$ by design. Rearrangement of (II-4) yields

$$(\gamma_i^{t+1} + 1)^2 \geq (\gamma_i^{t+1} - 1)^2 + (2\|z_{i,N_I}^{t+1}\|_2)^2, \quad (II-5)$$

which is equivalent to the standard second-order conic constraint

$$\|\gamma_i^{t+1} + 1\|_2^2 \geq \|[\gamma_i^{t+1} - 1, 2z_{i,N_I}^{t+1}]^T\|_2^2. \quad (II-6)$$

III. BATTERY CELL PARAMETERS AND ESTIMATED HEAT CAPACITY AND THERMAL RESISTANCE

For numerical case studies presented in Section V of the main paper, each EV is equipped with 444 NMC-18650 battery cells depicted in Fig. 1, in the 2P5S configuration. Each NMC-18650 battery cell has capacity 3.3 Ah, nominal voltage $v_{flat,i} = 3.8$ V with operating voltage ranging from 3.3 V to 4.2 V, and maximum C-rate of 2.5 to avoid excessively sharp charge-discharge cycling. The initial coolant's temperature is assumed to be 298.5 K = 25.4°C. The fluid speed of coolant is 0.5 m/s. The battery density is 2055.2 [kg/m³], and battery heat capacity is 1169.1 [J/(kg·K)], battery thermal conductivity in the radius direction is 1.1433 [W/(m·K)] and in the angular direction is

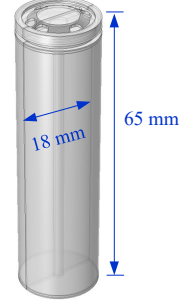


Fig. 1. An NMC-18650 battery cell.

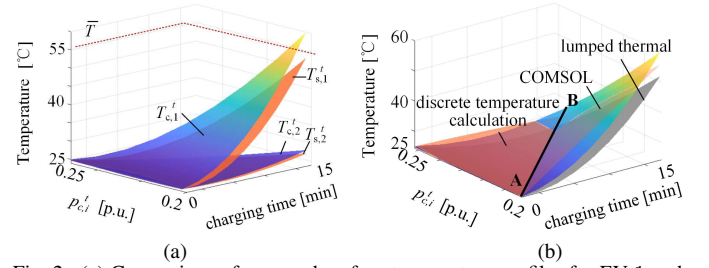


Fig. 2. (a) Comparison of core and surface temperature profiles for EV 1 and EV 2. (b) core temperature profiles of EV 1 comparing COMSOL-based DT simulation, discrete-time temperature calculations from (5), and the lumped thermal model used by M_2 [13].

35.875 [W/(m·K)]. Other parameters related to electrochemical and thermal dynamics are provided in Table I.

Parameters $C_{c,i}$, $C_{s,i}$, $R_{c,i}$, and $R_{s,i}$ are estimated under the nominal current for the NMC-18650 battery cell using the least-squares error algorithm from simulated datasets obtained under no external cooling, forced air, water cooling, and the double-sided cooling of ethylene glycol, all with $T_{amb,i} = 25^\circ\text{C}$. The resulting values are reported in Table II. In the COMSOL simulations, the speeds of water, forced air and ethylene glycol are selected as 0.5 m/s, the coolant's temperature is 25°C . The heat transfer coefficients of no external cooling, forced air, water cooling, and the double-sided cooling of ethylene glycol are set to 5 W/(m²·K), 10 W/(m²·K), 10 W/(m²·K), and 10 W/(m²·K), respectively. Using the parameter values in Table II, we find that the maximum errors between the COMSOL simulations and calculated $T_{c,i}^t$ and $T_{s,i}^t$ are 2.0°C , 1.6°C , 0.6°C , and 0.6°C under the no external cooling, forced air, water cooling, and the double-sided cooling of ethylene glycol, respectively.

IV. ADDITIONAL VALIDATION OF THERMAL MODEL IN (5)

We obtain values of $T_{c,i}^t$ and $T_{s,i}^t$ over $\Delta t = 15$ minutes from the initial SoC $S_i^t = 0.1$. For illustration, we plot in Fig. 2(a) the core temperatures $T_{c,1}^t$ and $T_{c,2}^t$, surface temperatures $T_{s,1}^t$ and $T_{s,2}^t$ from COMSOL-based digital twin (DT) simulation for EV 1 and EV 2. Visual examination of these surfaces offers further substantiation that monitoring surface temperatures alone may leave TR risk undetected, especially for batteries with poorer cooling capabilities as is the case for EV 1.

To assess the effectiveness of the thermal model in (5) as compared with the lumped thermal model used in M_2 [13], we select EV 1 and plot pertinent core temperature profiles in Fig. 2(b). The maximum error between COMSOL-based DT simulation and discrete-time temperature calculations yielded from (5) is 1.9°C . In contrast, the lumped thermal model [13] yields the surface plotted in grey colour is

TABLE I
PARAMETERS FOR ELECTROCHEMICAL AND THERMAL KINETICS

Parameter	Negative electrode	Separator	Positive electrode
Electrode plate area [m ²]	0.163	0.163	0.163
Electrode thickness [m]	$78 \cdot 10^{-6}$	$20 \cdot 10^{-6}$	$45 \cdot 10^{-6}$
Li ⁺ diffusion coefficient [m ² /s]	$3.9 \cdot 10^{-5}$	-	$1.8 \cdot 10^{-8}$
Active electrode volume fraction [%]	0.6	-	0.6
Electrolyte phase volume fraction [%]	0.3	-	0.3
Maximum solid phase concentration [mol/m ³]	31507	-	49000
Particle radius [m]	$6 \cdot 10^{-6}$	-	$5 \cdot 10^{-6}$
Reaction rate efficiency [A/m ²]	$9.77 \cdot 10^{-2}$	-	$1.19 \cdot 10^{-2}$
Exchange current density of side reaction [A/m ²]	10	-	10
Initial electrolyte concentration [mol/m ³]	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$
Binder volume fraction [%]	0.1	-	0.1
Separator volume fraction [%]	-	0.4	-
Thickness of current collectors [m]	$10 \cdot 10^{-6}$	-	$10 \cdot 10^{-6}$
Thermal conductivity [W/(m·K)]	1.04	0.344	1.58
Electrode density [kg/m ³]	1347.33	1008.98	2328.5
Electrode heat capacity [J/(kg·K)]	1437.4	1978.2	1269.2
Current collector thermal conductivity [W/(m·K)]	398	-	170
Current collector density [kg/m ³]	8933	-	2770
Current collector heat capacity [J/(kg·K)]	385	-	875

TABLE II
PARAMETERS OF HEAT CAPACITY AND THERMAL RESISTANCE

Coolant	$C_{c,i}$ [J/K]	$C_{s,i}$ [J/K]	$R_{c,i}$ [K/W]	$R_{s,i}$ [K/W]
None	284.1	0.0015	260.2	3663
Forced air	174.9	0.0017	253.9	4166
Water	3.3	6.3	8446.9	1040.2
Ethylene glycol	3.1	6.1	8447	1042

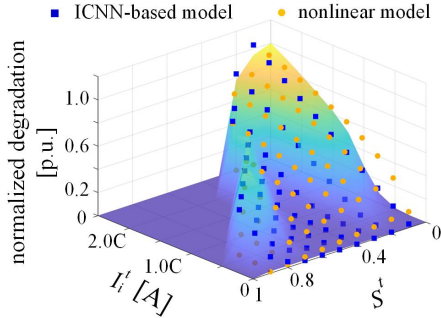


Fig. 3. Comparison between nonlinear and ICNN-based degradation models.

less accurate, differing from the COMSOL-based DT simulation with a maximum error of 4.5°C. This limitation arises because the lumped thermal model assumes uniform temperature distribution throughout the cell and thus does not account for the potential thermal security in the core.

V. LINEAR BATTERY DEGRADATION FORMULATION

In [15], the linear cost function for battery degradation F_i^t is defined for each charging and discharging cycle at time t until the required state of charge (SoC) in the last time period $t = N_t$, i.e., $S_i^{N_t} = \bar{S}_i$, yielding

$$F_i^t = C_i \frac{\kappa_i}{100} \left(-\frac{1}{\eta_{v,i}} p_{d,i}^t + \bar{S}_i - S_i^0 \right) \quad (\text{V-7})$$

where κ_i denotes the slope of the linear approximation of the EV i 's battery life as a function of cycles, C_i [\$/Ah] represents the purchase cost of EV i 's battery normalized by its useful capacity, S_i^0 is the initial SoC value, and $1/\eta_{v,i}$ represents the discharging efficiency of EV i .

VI. COMPARISON WITH NONLINEAR DEGRADATION MODEL

In [17], the nonlinear cost function for battery degradation G_i^t is approximated from experimental data with respect to depth of discharge (DoD) $1 - S_i^t$ and cell current I_i^t (quantified by the C-rate). The cost function is expressed as

$$G_i^t = \frac{1}{f_c(I_i^t) \cdot f_d(S_i^t)}, \quad (\text{VI-8})$$

where f_c and f_d represent battery degradation cost due to DoDs and cell currents, respectively, and they are approximated as

$$f_d = \begin{cases} 4000, & \text{if } S_i^t > 0.95, \\ 946.1 \cdot (1 - S_i^t)^{-1.079}, & \text{if } S_i^t \leq 0.95, \end{cases} \quad (\text{VI-9a})$$

$$f_c = \begin{cases} 4, & \text{if } I_i^t < 0.2, \\ 1.041 \cdot (I_i^t)^{-0.445}, & \text{if } 0.2 \leq I_i^t \leq 2.5. \end{cases} \quad (\text{VI-9b})$$

We compare (VI-8) and the ICNN-based formulation in (9b)–(9d) under a random set of 66 SoC values. We normalize the electrochemical degradation costs in a continuous range from 0 to 1 and overlay solutions from both models on the PDE-based degradation surface $\Delta\theta_1^{t+1}$ in Fig. 3. Among data points plotted in Fig. 3, 84.8% of markers have less than 1% error for the ICNN-based model compared with 19.7% for the nonlinear formulation. Furthermore, the ICNN-based model has less than 2% error across the entire domain plotted, demonstrating superior accuracy for the optimal V2G dispatch.